

Data and Code for: “A Machine Learning Approach to Analyze and Support Anti-Corruption Policy”

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This document provides comprehensive documentation for the replication kit associated with the paper titled '**A Machine Learning Approach to Analyze and Support Anti-Corruption Policy**'. It includes a detailed data availability statement outlining the sources and accessibility of all datasets used, along with step-by-step instructions for replicating the analyses and results presented in the paper.

1. Data Availability and Provenance Statement

Statement about Rights

The authors confirm that they have legitimate access and appropriate authorization to use the data presented in this manuscript.

Summary of Availability

The data are publicly available online through institutional sources or derived from datasets used in previously published papers.

Details on each Data Source

The project combines different data sources, outlined below.

1. Municipal corruption data are the ones generated and used by *Brollo et al. (2013)* and *Avis et al. (2018)*. Both papers based their measure on audit reports and data provided by the CGU.
2. Local finance data comes from the Finance Ministry online database (<https://www.tesourotransparente.gov.br/publicacoes/finbra-dados-contabeis-dos-municipios-1989-a-2012/2012/26>).
3. The general demographic and economic composition of each municipality comes from the 2000 Brazilian decennial census. Data can be openly accessed via the census website.

(<https://www.ibge.gov.br/en/statistics/social/population/22836-2022-census-3.html?=&t=downloads>)

4. Political affiliation and electoral outcomes of mayors are provided by the *Tribunal Superior Eleitoral*

(e.g., <https://www.tse.jus.br/eleicoes/eleicoes-anteriores/eleicoes-2004/candidaturas-votacao-e-resultados/resultados-das-eleicoes-2004>)

5. Municipal audit reports are published on the CGU website. We programmatically downloaded the entire library of audit reports for our time period as PDF files.

(<https://eaud.cgu.gov.br/relatorios?colunaOrdenacao=dataPublicacao&direcaoOrdenacao=DESC&tamanhoPagina=15&offset=0&fixos=#lista>, the scraped data are available here <https://doi.org/10.5281/zenodo.15129902>)

2. Replication Instructions

Software Requirements

- Stata (code was last run with version 16)

binscatter

binsreg

coefplot

csdid

ranktest

ivreg2

parmest

reghdfe

ftools

estout

drdid

- Python

Due to the large number of specific dependencies, we do not list them individually here. Instead, please refer directly to the *requirements.txt* file and Pipfile, which specify the required package versions. You can install Pipenv using: *pip install pipenv*. Then, use Pipenv to manage and install dependencies.

Calling the do file *code_stata/01_master.do* ensure that Stata calls the programs from the correct directory.

Computational Requirements

The Python code was run on a Linux Machine (Ubuntu 22.04) with 64GB of RAM and an Intel 12700k CPU. 32GB of RAM probably also suffices.

For Stata the code was last run on Intel Core i7 quad-core CPU 2,2 GHz , with 16 GB of RAM, macOS version 12.1

Time Requirements

Approximate time needed to reproduce the analyses on our server: 1 week if processed sequentially. Most of the time is needed to generate the predictions.

Replication Folder Structure

The replication folder is organized in the following subdirectories:

- **replication_package**
 - **code_python/**: Includes all Python scripts necessary to replicate figures and tables generated using Python. This directory also contains scripts for generating the essential intermediate datasets used in the subsequent Stata analyses.
 - **code_stata/**: Includes all Stata scripts necessary to replicate figures and tables generated using Stata.
 - **data/**: Contains all data files organized into the following subdirectories:
 - **data/source/**: Stores all raw, original data files.
 - **data/working/**: Initially empty; used to store intermediate files created during data cleaning and processing.
 - **data/final/**: Initially empty; used to store final datasets employed in analyses.
 - **output/**: Initially empty; designated for storing outputs such as figures, tables, and other results generated by both Python and Stata scripts. Sometimes Tables and Figures in the paper are manually edited from these outputs.

Instructions to Replicators

The provided code allows you to replicate all outputs, starting from raw data. Follow these steps to perform the replication:

1. **Setup:**
 - Navigate to the root folder via the command line.
 - Activate the virtual environment.
2. **Configuration:**
 - Define the appropriate file paths in:
 - `code_stata/01_master.do`
 - `code_stata/02_master.do`
3. **Execution:**
 - Run the `main.sh` script. This bash script automates the execution of the required Python and Stata scripts in the correct sequence, thereby generating all outputs.

The detailed workflow executed by `main.sh` includes the following steps:

Step 1: Generate Initial Datasets

- Generate datasets required for prediction tasks using Stata:

```
do "code_stata/01_master.do"
```

Step 2: Generate Predictions and first downstream analysis

- Execute all machine learning predictions, followed by preliminary downstream analyses in Python.
- Ensure directories matching `data/working/prediction_XYZ` do not exist, as they can cause execution failures.
- After generating predictions, proceed with downstream Python analyses.

Step 3: Generate Stata-based Outputs

- Produce the final set of results using Stata:

```
do "code_stata/02_master.do"
```

List of Tables and Figures

code	exhibit	dataset	notes
Table1.do	Table 1	dataset_AnalysisAudit.dta panel_budget_corruption.dta biscatter_AuditReports.dta	Stata
generate_plots.py	Table 2		Python

Table3.do	Table 3	dataset_AnalysisTransfer.dta	Stata
policy_analysis.py	Table 4		Python
	Table A1		No code. The table was constructed by identifying and aggregating the frequency of variables from each category-year combination appearing among the features selected by the XGBoost model. The total number of audits per year was calculated based on the data available from the single lottery selection, accessible at: https://www.gov.br/cgu/pt-br/assuntos/auditoria-e-fiscalizacao/programa-de-fiscalizacao-em-entes-federativos/edicoes-anteriores/municipios?b_start:int=20
hyperparam_selection.py	Table A2		Python
generate_plots.py	Table A3		Python
generate_plots.py	Table B4		Python
table_c5.py	Table C5		Python
See README in text_audit	Table C6		Python
	Table D7		No code
TableD8.do	Table D8	dataset_AnalysisTransfer.dta	Stata
TableD9.do	Table D9	dataset_AnalysisTransfer.dta	Stata
TableD10.do	Table D10	dataset_AnalysisTransfer.dta	Stata

TableD11_panel_A.do TableD11_panels_B_C.do	Table D11	dataset_AnalysisTransfer.dta predict_fold_0/999.dta estimates_2_1_fold_0/999.dta estimates_3_1_fold_0/999.dta estimates_2_2_fold_0/999.dta estimates_3_2_fold_0/999.dta estimates_2_3_fold_0/999.dta estimates_3_3_fold_0/999.dta	Stata
TableE12.do	Table E12	dataset_AnalysisAudit.dta	Stata
TableE13_columns_1_3.do TableE13_columns_4_9.do	Table E13	predict_fold_0/999_YearVersion.dta estimates_EventStudy_1_fold_0/999.dta estimates_EventStudy_2_fold_0/999.dta estimates_EventStudy_3_fold_0/999.dta	Stata
policy_analysis.py	Figure 1		Python
Figure2.do	Figure 2	biscatter_AuditReports.dta	Stata
	Figure 3	data/working/for_maps data/source/55mu500gc	The map file, located at data\source\maps.qgz , was generated using QGIS by joining data from data/working/for_maps (a file generated by the code) with the provided shapefile located at data/source/map/55mu500gc . Please ensure that the folder paths are configured correctly to replicate.

Figure4.do	Figure 4	dataset_AnalysisAudi.dta	Stata
policy_analysis.py	Figure 5		Python
policy_analysis.py	Figure A1		Python
FigureA2.do	Figure A2	dataset_AnalysisAudi.dta panel_budget_corruption.dta	Stata
FigureB3.do	Figure B3	dataset_AnalysisAudi.dta corruption_predictions_NS_b_Avis_foldAll.dta	Stata
local_interpretability.py	Figure C4		Python
FigureE5.do	Figure E5	dataset_AnalysisAudi.dta	Stata
FigureE6.do	Figure E6	dataset_AnalysisAudi.dta panel_budget_corruption.dta	Stata
FigureE7.do	Figure E7	dataset_AnalysisAudi.dta panel_budget_corruption.dta	Stata
FigureE8.do	Figure E8	dataset_AnalysisAudi.dta victory_margin_elections.dta	Stata
	Figure F9		Python
	Figure F10		Python
	Figure G14		No code

List of Original Datasets Source

dataset	source
ativo_1998.xls, ativo_1999.xls, ativo_2000.xls, ativo_2001.xls, ativo_2002.xls, ativo_2003.xls,	https://www.tesourotransparente.gov.br/publicacoes/finbra-dados-contabeis-dos-municipios-19

ativo_2004.xls, ativo_2005.xls, ativo_2006.xls, ativo_2007.xls, ativo_2008.xls, ativo_2009.xls, ativo_2010.xls, ativo_2011.xls, ativo_2012.xls, despesa_1998.xls, despesa_1999.xls, despesa_2000.xls, despesa_2001.xls, despesa_2002.xls, despesa_2003.xls, despesa_2004.xls, despesa_2005.xls, despesa_2006.xls, despesa_2007.xls, despesa_2008.xls, despesa_2009.xls, despesa_2010.xls, despesa_2011.xls, despesa_2012.xls, DespFuncao_2004.xls, DespFuncao_2005.xls, DespFuncao_2006.xls, DespFuncao_2007.xls, DespFuncao_2008.xls, DespFuncao_2009.xls, DespFuncao_2010.xls, DespFuncao_2011.xls, DespFuncao_2012.xls, passivo_1998.xls, passivo_1999.xls, passivo_2000.xls, passivo_2001.xls, passivo_2002.xls, passivo_2003.xls, passivo_2004.xls, passivo_2005.xls, passivo_2006.xls, passivo_2007.xls, passivo_2008.xls, passivo_2009.xls, passivo_2010.xls, passivo_2011.xls, passivo_2012.xls, receita_1998.xls, receita_1999.xls, receita_2000.xls, receita_2001.xls, receita_2002.xls, receita_2003.xls, receita_2004.xls, receita_2005.xls, receita_2006.xls, receita_2007.xls, receita_2008.xls, receita_2009.xls, receita_2010.xls, receita_2011.xls, receita_2012.xls	89-a-2012/2012/26
demographic_controls.dta	Information is based on the publicly available census data: https://www.ibge.gov.br/en/statistics/social/population/22836-2022-census-3.html?=&t=downloads This dataset was provided by F. Cavalcanti (francisco.dlcavalcanti@ufpe.br) and are also available from the data package of Boffa et al 2024. The variable we collect are Mean income, Agriculture (% employed), Industry (% employed), Commerce (% employed), Transport (% employed), Service (% employed), Public administration (% employed), Employed population, Graduated people, Poor population, Gini coefficient
corruptiondata_lotteries_brollo.dta	This dataset was downloaded from Fernanda Brollo personal website in 2016 (https://sites.google.com/site/fernandabrollo). The dataset is not available anymore. The same data have been used for a published paper Brollo et al 2013. The replication pack of Brollo et al 2013 provides a cleaned .dta that is

	based on the original dataset.
corruptiondata_lotteries_avis.dta	This dataset was provided by Eric Avis (eric.avis@hec.ca). The same data have been used for a published paper Avis et al 2018. The replication pack of Avis et al 2018 provides a cleaned .dta that is based on the original dataset.
info_lotteries.dta	The dataset was created by the authors using information collected from the CGU website about the dates of the different lotteries.
text_files.zip	Contains text files of audit report and are available from the CGU website. We scraped the content in 2020 using the <code>scrape_pdfs.py</code> . The structure of the website is changed and the scraper does not work anymore. All files can be anyway manually retrieved from the websites. The file needs to be downloaded from https://zenodo.org/records/15129902 and included in the repository.
transferências_para_municípios_yearly.csv	This dataset was downloaded from the website of the Brazilian national treasury (https://www.tesourotransparente.gov.br/consultas/transferencias-constitucionais-realizadas).
POP-2000.xls, POP-2001.xls, POP-2002.xls, POP-2003.xls, POP-2004.xls, POP-2005.xls, POP-2006.xls, POP-2008.xls, POP-2009.xls	This datasets were downloaded from the website of the Brazilian institute of Geography and Statistics (IBGE) (https://www.ibge.gov.br/en/statistics/downloads-statistics.html).
votacao_candidato_munzona_2000_BRASIL.csv votacao_candidato_munzona_2004_BRASIL.csv votacao_candidato_munzona_2008_BRASIL.csv	These datasets were downloaded from the website of the Superior Electoral Tribunal (TSE) (https://dadosabertos.tse.jus.br/dataset/resultados-2000 ; https://dadosabertos.tse.jus.br/dataset/resultados-2004 ; https://dadosabertos.tse.jus.br/dataset/resultados-2008).
municipios_brasileiros_tse.csv	This dataset was downloaded from this GitHub portal (https://github.com/betafcc/Municipios-Brasileiros-TSE/blob/master/municipios_brasileiros_tse.csv).
55mu500gc	This is the shape file to generate the maps https://geoftp.ibge.gov.br/organizacao_do_territorio/malhas_territoriais/malhas_municipais/municipio_2005/escala_500mil/proj_geografica/arcview_shp/brasil/

3. References

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